AI-based Anomaly Detection with Sensor Data & AWS IoT Core

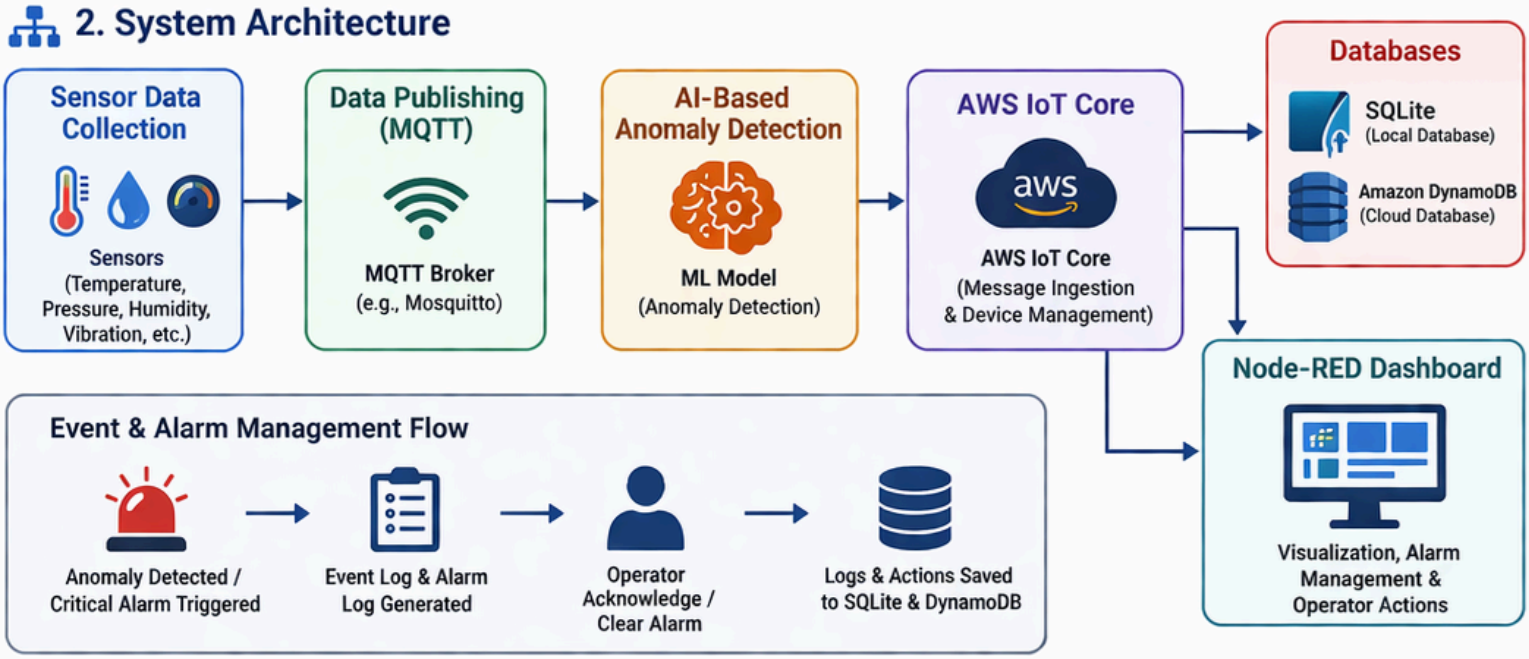
1. Project Overview

This project demonstrates real-time AI-based anomaly detection using sensor data. It integrates AI algorithms for anomaly prediction, MQTT for sensor data streaming, AWS IoT for cloud communication, Node-RED for visualization, and SQLite/ DynamoDB for local and cloud database storage. The system captures event logs and alarm logs when anomalies are detected and critical alarms are triggered. Operators can acknowledge or clear alarms to improve process efficiency. All event, alarm, and operator actions are saved for reference in both the local database and the cloud database.

Key Features

- Real-time Sensor Data Collection and Publishing
- AI-based Anomaly Detection
- Integration with AWS IoT for Cloud Monitoring
- Dashboard-ready Data Outputs for Visualization

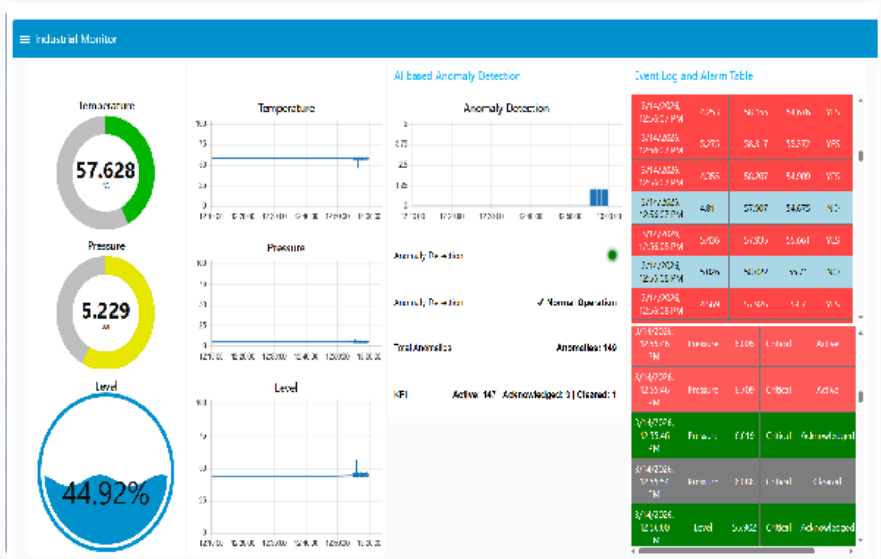
2. System Architecture



3. Workflow

1. **Sensor Data Collection** – Sensors collect data in real time.
2. **Data Publishing** – Sensor data is published to an MQTT broker.
3. **AI Anomaly Detection** – ML model analyzes data and detects anomalies.
4. **AWS IoT Integration** – Data and alerts are sent to AWS IoT Core.
5. **Logging** – Events and alarms are stored in SQLite (local) and DynamoDB (cloud).
6. **Visualization** – Node-RED dashboard displays real-time data and alarms.
7. **Operator Actions** – Alarms can be acknowledged or cleared; actions are logged for reference.

4. Dashboard Overview



AI-Based IIoT Tank Monitoring & Anomaly Detection

Developed an end-to-end IIoT system for real-time monitoring and AI-driven anomaly detection of industrial tank parameters.

Role: Designed & developed full IIoT system

Problem

- ✓ Process instability due to abnormal temperature, pressure, and level readings
- ✓ Unexpected downtime caused by undetected anomalies

Approach

- ✓ Simulated sensor data using Python
- ✓ Streamed real-time data via MQTT (Mosquitto)
- ✓ Applied Isolation Forest for anomaly detection

Solution

- ✓ Integrated Node-RED dashboards for real-time monitoring
- ✓ Implemented alert system for abnormal behavior
- ✓ Cloud-ready architecture for scalable deployment

Impact

- ✓ 15–20% defect reduction
- ✓ 20–50% downtime reduction (predictive maintenance)
- ✓ 8–20% productivity improvement
- ✓ 10–25% energy savings